Vertiport Planning in Chicago

Mathematical model, working document v1.1[1]

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Abstract

This document describes the optimization model of the project. The model builds on Wang, Jacquillat and Vaze (2022), "Vertiport Planning for Urban Aerial Mobility: An Adaptive Discretization Approach," *M&SOM* 24(6):3215–3235. Their framework combines facility location with demand that depends on travel times. It also uses a distributionally robust demand function. Below, "the paper" refers to this article. This project uses a simplified version without the operations layer. It asks two questions.

RQ1: What changes when customer expectations are estimated for each origin–destination pair from observed rideshare fares and times, instead of being the same for all?

RQ2: How robust is the network to demand at different times of day? What does a network planned for a single period lose over a year?

1 Overview

The model has two layers. The *precomputation* layer (Python) converts trip records into travel times and switching shares for every possible routing. All nonlinear relations are computed there. The *optimization* layer is a mixed-integer linear program (MILP) solved with Gurobi. It chooses the vertiport sites and routes each origin–destination flow. All reported runs maximize the number of air-taxi passengers. The profit version uses the margins of Section 4.5 as weights. Figure 1 shows the pipeline.

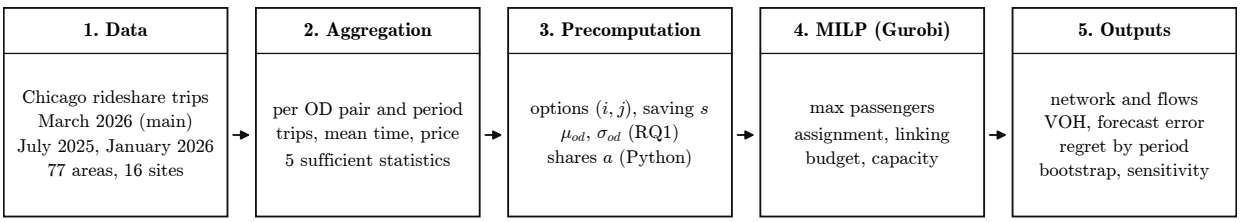


Figure 1: From trip records to the network and the metrics of Section 9. s is the time saving of an option, a the share of travelers who switch, and μ_{od}, σ_{od} the mean and standard deviation of the required saving on OD pair (o, d). VOH is the value of heterogeneity (Definition 1).

Figure 2 shows the choice for one origin–destination (OD) pair in one period. Each traveler either stays on the ground (rideshare) or flies through an open vertiport pair (i, j).

2 Sets and indices

Table 1 lists the sets. The index t always means a period of the day. Times are written with a capital T.

[1]Version 1 was written before the Chicago runs. Version 1.1 matches the code and the reported results.

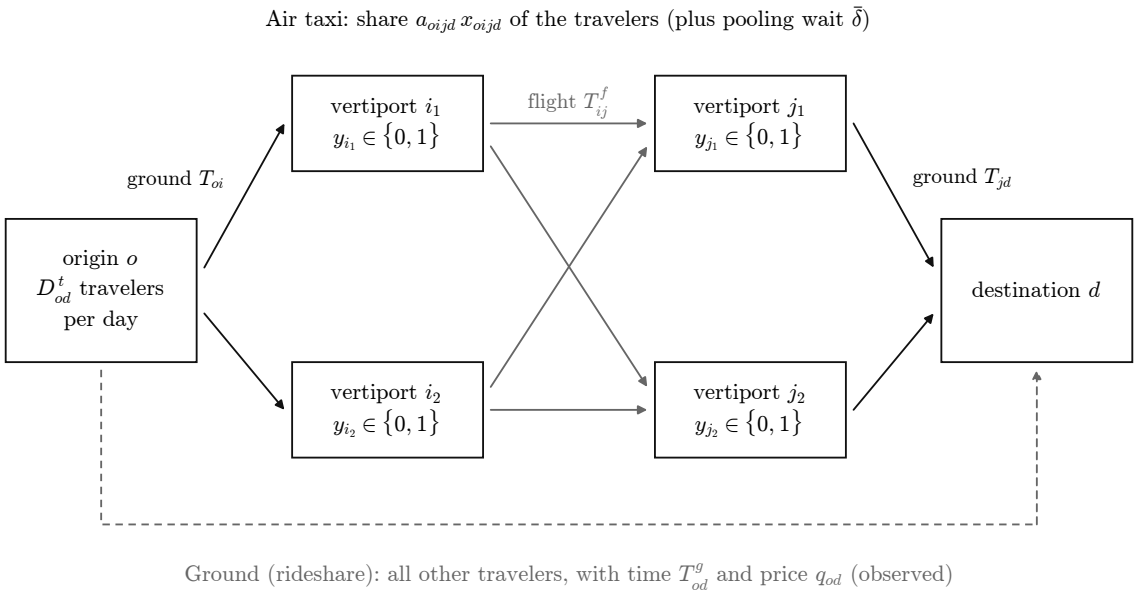


Figure 2: Routing options for OD pair (o, d) in period t. Black arrows are ground legs to and from vertiports. Grey arrows are flight legs. The dashed arrow shows the travelers who stay on the ground.

Table 1: Sets and indices.

Set	Index	Meaning
$\mathcal{Z}$	o, d	Zones: the 77 Chicago community areas. A trip starts in o and ends in d
$\mathcal{V} \subseteq \mathcal{Z}$	i, j	Candidate sites for a vertiport. Main runs: 16 sites (O'Hare, Midway and Vertiport Chicago at their exact locations, five high-demand areas, eight areas with cluster centers of Volakakis and Mahmassani 2024). Robustness runs: all 77 areas
$\mathcal{T}$	t	Periods $\{\text{am}, \text{mid}, \text{pm}, \text{oth}, \text{we}\}$: weekday 7–10, 10–16, 16–19, other weekday hours, and weekend
$\mathcal{P} \subseteq \mathcal{Z} \times \mathcal{Z}$	(o, d)	OD pairs kept in each period: $o \neq d$, mean ground time of at least 20 minutes, at least one trip per day
$\mathcal{K}^t_{od} \subseteq \mathcal{V} \times \mathcal{V}$	(i, j)	Useful vertiport pairs for (o, d) in period t: travelers board at i and land at j (Section 4.2)

3 Data and parameters

3.1 Observed in the Chicago rideshare data

The data are the Transportation Network Providers trips of the City of Chicago (Socrata dataset `6dvr-xwnh`). The main results use March 2026. The seasonal check uses July 2025 and January 2026. Trips are aggregated by pickup area, drop-off area and period, one day at a time. Shared trips are removed. Trips under 60 seconds or over 3 hours are removed. Trips without a fare or without an area are removed. Ground time is the field `trip_seconds`. Its timestamps are rounded to 15 minutes. Ground price is `fare` + `additional_charges`. Fares are rounded to \$2.50. Table 2 lists the observed quantities.

Table 2: Quantities observed in the data.

Symbol	Meaning	Unit
D^t_{od}	Number of rideshare trips from o to d, per day of period type t	trips/day
$T^{g,t}_{od}$	Average ground (rideshare) time from o to d in period t	min
q_k, T_k	Price and ground time of one trip k	\$, min
$m^{(1..5),t}_{od}$	Five averages over the trips of (o, d, t), used to estimate expectations (Section 5)	various
ℓ_{od}, ℓ_{ij}	Straight-line distance between two zone centers, or between two sites	km

3.2 Assumed parameters

The assumed values come from public sources. Table 3 gives each default, a plausible range, and the values actually tested.

Table 3: Assumed parameters. "Via h": this fixed time adds minutes to the air-taxi trip in the same way as h, so the tests of h cover it (Section 10.4).

Symbol	Meaning	Default	Plausible	Tested
v	Cruise speed of the aircraft (240 km/h for NYC)	200 km/h	150–240	no
T_{oh}	Fixed time per flight (boarding, takeoff, landing)	10 min	6–20	via h
$\bar{\delta}$	Wait for other passengers to fill the aircraft	5 min	2–10	via h
T_0	Ground time when the trip starts or ends in the vertiport's own zone	5 min	3–8	no
h	Extra minutes for the hassle of changing vehicles	0 min	0–15	0, 5, 10, 15
$s_{\min}$	Smallest time saving for which an option is offered	0.15	0–0.15	0, 0.15
$T_{\max}$	Longest allowed ground leg to or from a vertiport	25 min	15–30	no
k	Number of nearest vertiports considered at each end of a trip	3	3–8	3, 8
μ, σ	Expectations that are the same for all travelers (NYC only)	0.40, 0.20	–	μ: 0.33–0.40
P_0, r, b_0	Base fare, fare per mile, median value of time	Table 4	–	4 scenarios
CV_b	How much the value of time varies across travelers	0.84	0.5–1.2	no
κ, c^g, g_i	Flight cost per km, ground-leg cost, daily cost of a site (profit version only)	not used	–	–
w_t	Days per year of period type t	250 / 115	–	–

Table 4 gives the four fare scenarios. Each scenario pairs a fare level with the travelers who would pay it.

Table 4: Fare and value-of-time scenarios (2024 dollars). The fare per mile applies to the straight-line OD distance. The value of time is lognormal, with median b_0 and $\mathrm{CV}_b = 0.84$.

Scenario	P_0	r **(/mile)**	b_0 **(/hour)**	**Basis**
Launch	\$15	\$6.00	\$62.00	near-term fare estimates; airport and business travelers
Launch, pessimistic	\$15	\$12.00	\$62.00	current helicopter shuttle prices
Adopter (main)	\$15	\$3.50	\$31.00	operator targets; air-taxi survey of Boddupalli et al. (2024)
Mature	\$15	\$2.50	\$20.10	long-run fare estimates; USDOT personal local travel

4 Precomputation

4.1 Ground and flight times

Ground legs. The ground time between zones u and v in period t is

$$T^t_{uv} = \begin{cases} T_0 & \text{if } u = v, \\ \text{observed average } T^{g,t}_{uv} & \text{if the pair has at least 2 trips per day,} \\ \alpha^t + \beta^t \, \ell_{uv} & \text{otherwise.} \end{cases} \tag{1}$$

The line $\alpha^t + \beta^t \, \ell_{uv}$ is fitted on the pairs with at least 5 trips per day.

Flight legs. The flight time from site i to site j is the distance divided by the speed, plus the fixed time per flight:

$$T^f_{ij} = \frac{60 \, \ell_{ij}}{v} + T_{\mathrm{oh}}. \tag{2}$$

4.2 Useful vertiport pairs

Let $N^{\mathrm{acc},t}(o)$ be the k sites closest to o by ground time, with $T^t_{oi} \le T_{\max}$. Let $N^{\mathrm{egr},t}(d)$ be the k sites closest to d, with $T^t_{jd} \le T_{\max}$. A traveler can board at any site in the first set and land at any site in the second set:

$$\mathcal{K}^t_{od} = \left\{ (i,j) \in N^{\mathrm{acc},t}(o) \times N^{\mathrm{egr},t}(d) \ : \ i \neq j \right\}. \tag{3}$$

4.3 Air-taxi trip time and time saving

The air-taxi trip has five parts: the ground leg to the vertiport, the pooling wait, the flight, the ground leg from the vertiport, and the transfer penalty. The trip time follows equation (5) of the paper, but with a fixed pooling wait:

$$\tau^t_{oijd} = T^t_{oi} + \bar{\delta} + T^f_{ij} + T^t_{jd} + h, \tag{4}$$

$$s^t_{oijd} = 1 - \frac{\tau^t_{oijd}}{T^{g,t}_{od}}. \tag{5}$$

The saving s is the share of the ground time that the air taxi saves. For example, $s = 0.3$ means the air taxi is 30% faster than the rideshare.

The transfer penalty h is an extra cost for changing vehicles, written in minutes. The utility of Section 4.4 has no such cost, and neither do equations (19)–(20) of the paper. The paper's

expectations come from surveys, which include this cost implicitly. Expectations from trip data do not include it. The default is $h = 0$. Section 10.4 tests larger values. Figure 3 shows the parts of the trip for three NYC examples.

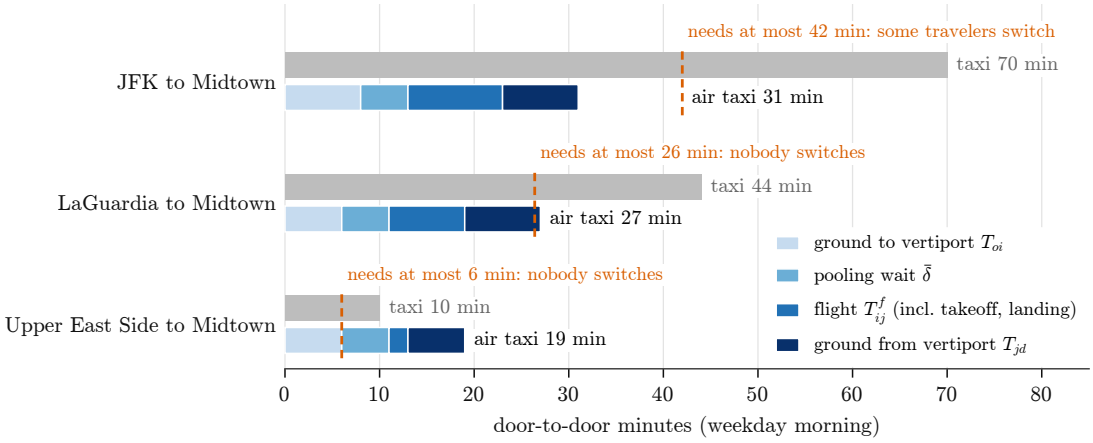


Figure 3: Air-taxi trip versus taxi in the NYC validation. Taxi times are observed. Air-taxi components are illustrative. The dashed line is the $\mu = 0.40$ threshold. Only long trips still save time after the fixed components.

4.4 Demand: the share of travelers who switch

As in Section 4.4 of the paper, a traveler on (o, d) compares two utilities: $V - P_{od} - b\,\tau$ for the air taxi and $V - q_{od} - b\,T^g_{od}$ for the ground. Here V is the value of the trip, P_{od} and q_{od} are the two prices, and b is the traveler's value of time. She flies if and only if

$$\tau \le (1-\theta)\,T^g_{od} \quad \Longleftrightarrow \quad \theta \le s, \qquad \text{where} \quad \theta = \frac{P_{od} - q_{od}}{b\,T^g_{od}}. \tag{6}$$

θ is the *required time saving*: the saving a traveler needs before flying is worth its extra price. It is the extra price, converted to minutes with b, and divided by the ground time. With times in minutes, b is in dollars per minute.

θ differs across travelers and trips. Its mean is μ and its standard deviation is σ. The share of travelers who fly is $\Pr(\theta \le s)$. We use two versions of this share. The first is the paper's distributionally robust (DRO) share. The second is an exact share under stated distributions. The exact share gives the main results.

DRO share (Proposition 1 of the paper). Suppose only μ and σ are known. The DRO share is the smallest share among all distributions with this mean and standard deviation. Cantelli's one-sided Chebyshev inequality gives it:

$$\inf_{F:\ \mathbb{E}_F\theta=\mu,\ \mathrm{Var}_F\theta=\sigma^2} \Pr_F(\theta \le s) = \begin{cases} \dfrac{(s-\mu)^2}{\sigma^2 + (s-\mu)^2} & s \ge \mu, \\ 0 & s < \mu. \end{cases} \tag{7}$$

With $\tilde{\tau} = (s-\mu)/\sigma$, the paper's share is

$$a^t_{oijd} = \frac{\tilde{\tau}^2}{1+\tilde{\tau}^2} \ \text{ if } \tilde{\tau} > 0, \qquad a^t_{oijd} = 0 \text{ otherwise}, \qquad \tilde{\tau} = \frac{s^t_{oijd} - \mu_{od}}{\sigma_{od}}. \tag{8}$$

Figure 4 plots this share. For NYC, $\mu_{od} = \mu$ and $\sigma_{od} = \sigma$ on all pairs. For Chicago, both are estimated from data (Section 5). The share is zero whenever $s \le \mu$. On most Chicago pairs $\mu_{od} > 1$, so the DRO share predicts almost no demand. For this reason, the main results use the

exact share. Both shares are computed before the optimization, so the model stays linear. In the paper, τ depends on pooling and detour decisions. The S-curve is therefore part of the model and requires adaptive discretization.

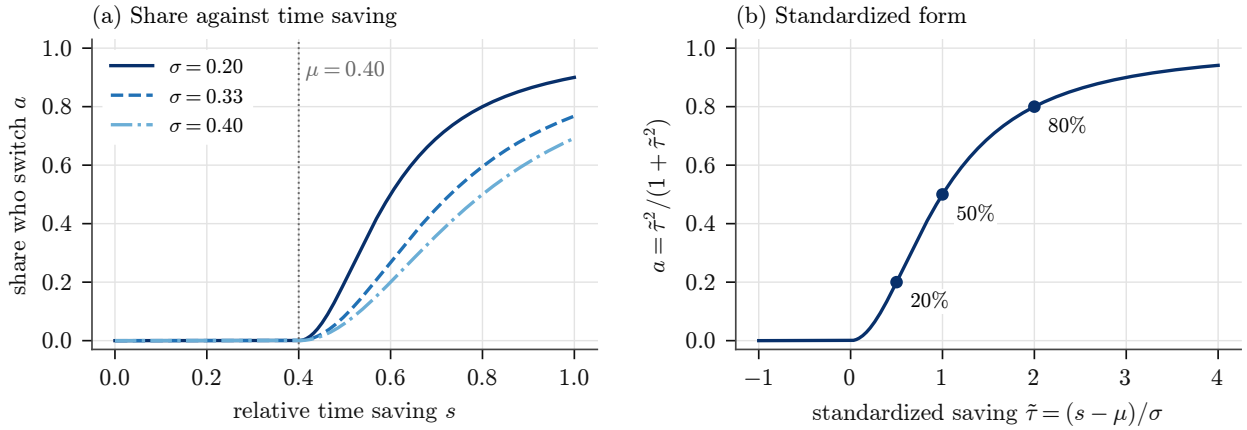


Figure 4: The DRO share (8). (a) Nobody switches until the saving exceeds μ. A larger σ makes the curve less steep. (b) In standardized units the curve is first convex, then concave. 20%, 50% and 80% of travelers switch at $\tilde{\tau} = 0.5$, 1 and 2.

Exact share (main results). Let $X = (P_{od} - q)/T$ be the extra price per minute of ground time. Then $\theta = X/b$. A traveler chooses an option with saving s if and only if $X \le s\,b$. We assume X is normal on each OD pair, with the mean and standard deviation of Section 5. We assume b is lognormal (median b_0, $\mathrm{CV}_b = 0.84$) and independent of X. The share is then the probability $\Pr(X \le s\,b)$, averaged over the values of time b:

$$a^t_{oijd} = \mathbb{E}_b\left[\Phi\left(\frac{s^t_{oijd}\,b - \mathbb{E}[X]}{\mathrm{sd}(X)}\right)\right]. \tag{9}$$

Φ is the standard normal distribution function. The average over b is computed numerically with 24-point Gauss–Hermite quadrature. An option is offered only if $s^t_{oijd} \ge s_{\min} = 0.15$. This excludes travelers who would switch only for price. This happens when surge pricing makes the ground fare higher than the air-taxi fare.

4.5 Fares and margins

The fare is door to door and grows with the straight-line OD distance. The rate r is per mile, and $0.621\,\ell_{od}$ converts kilometers to miles. The operator pays for the flight and both ground legs:

$$P_{od} = P_0 + r \cdot 0.621\,\ell_{od}, \tag{10}$$

$$c_{oijd} = \kappa\,\ell_{ij} + c^g\,(T^t_{oi} + T^t_{jd}), \tag{11}$$

$$\pi_{oijd} = P_{od} - c_{oijd}. \tag{12}$$

c_{oijd} is the cost per passenger: flight cost plus ground-leg cost. π_{oijd} is the margin per passenger. With $\pi_{oijd} = 1$ for all options, the objective counts passengers. All reported runs (NYC and Chicago) use this objective. The cost parameters κ, c^g and g_i are therefore not calibrated.

5 Customer expectations from trip data (RQ1)

For one trip k on (o, d) in period t, with price q_k and ground time T_k, the required time saving is

$$\theta_k = \frac{P_{od} - q_k}{b\,T_k}. \tag{13}$$

We need the mean μ_{od} and the standard deviation σ_{od} of θ on each OD pair.

Sufficient statistics. Downloading every trip would be slow. Instead, the City's data server returns five averages per (o, d, t):

$$m_1 = \mathbb{E}\left[\tfrac{1}{T}\right], \quad m_2 = \mathbb{E}\left[\tfrac{q}{T}\right], \quad m_3 = \mathbb{E}\left[\tfrac{1}{T^2}\right], \quad m_4 = \mathbb{E}\left[\tfrac{q}{T^2}\right], \quad m_5 = \mathbb{E}\left[\tfrac{q^2}{T^2}\right]. \tag{14}$$

These five numbers are enough for any air-taxi fare. For the extra price per minute $X = (P_{od} - q)/T$, the first two moments are

$$\mathbb{E}[X] = P_{od}\, m_1 - m_2, \qquad \mathbb{E}[X^2] = P_{od}^2\, m_3 - 2P_{od}\, m_4 + m_5. \tag{15}$$

Assume b is lognormal, with median b_0 and $\omega^2 = \ln(1 + \mathrm{CV}_b^2)$, and independent of the trip. Then $\mathbb{E}[1/b] = e^{\omega^2/2}/b_0$ and $\mathbb{E}[1/b^2] = e^{2\omega^2}/b_0^2$, so

$$\mu_{od} = \mathbb{E}[\theta] = \mathbb{E}[1/b]\,\mathbb{E}[X], \tag{16}$$

$$\sigma_{od}^2 = \mathbb{E}[1/b^2]\,\mathbb{E}[X^2] - \mu_{od}^2. \tag{17}$$

Fare rounding. Fares are rounded to the nearest \$2.50. This rounding adds noise to X that travelers do not see. Assume the rounding error ε is uniform on $[-1.25, 1.25]$ and independent of T. It then adds $\mathrm{Var}(\varepsilon)\, m_3$ to $\mathrm{Var}(X)$. We remove it:

$$\mathrm{Var}^{\mathrm{corr}}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 - \frac{2.5^2}{12}\, m_3, \qquad \sigma_{od}^{2,\mathrm{corr}} = \sigma_{od}^2 - \frac{2.5^2}{12}\,\mathbb{E}[1/b^2]\, m_3. \tag{18}$$

Thin OD pairs. Some OD pairs have few trips (n_{od} in the month). Their standard deviation is noisy. For them, $\mathrm{sd}(X)$ is moved toward η_t, the median of $\mathrm{sd}(X)$ over all OD pairs of period t:

$$\widehat{\mathrm{sd}}(X) = \frac{n_{od}\,\mathrm{sd}(X) + 30\,\eta_t}{n_{od} + 30}. \tag{19}$$

The minimum is \$0.01 per minute. This is a simple shrinkage rule, in the spirit of Efron and Morris (1975). The weight 30 is our choice.

Uniform benchmark (the "average customer"). The benchmark keeps the same travelers but removes the OD-specific information. For the exact share, every OD pair receives the trip-weighted mixture of all OD distributions:

$$a^{\mathrm{unif}}(s) = \frac{\sum_{od} D_{od}\, a_{od}^{\mathrm{exact}}(s)}{\sum_{od} D_{od}}. \tag{20}$$

For the DRO share, the pooled mean is $\bar{\mu} = \sum_{od} D_{od}\mu_{od} / \sum_{od} D_{od}$. By the law of total variance, the pooled variance is $\bar{\sigma}^2 = \sum_{od} D_{od}(\sigma_{od}^2 + \mu_{od}^2) / \sum_{od} D_{od} - \bar{\mu}^2$. Both models then have the same average level. They differ only in heterogeneity.

Scenarios. Realistic fares give $\theta > 1$ on many OD pairs. Results are therefore reported for the four scenarios of Table 4, not for one fare. Figure 5 shows how θ changes with the fare and the value of time.

6 Decision variables

The model makes two kinds of decisions: where to build, and how to route travelers.

$$y_i = \begin{cases} 1 & \text{if a vertiport is built in zone } i, \\ 0 & \text{otherwise,} \end{cases} \qquad i \in \mathcal{V},$$

$$x_{oijd}^t = \text{share of the travelers from } o \text{ to } d \text{ in period } t \text{ that is sent through vertiports } i \text{ then } j, \qquad (o,d) \in \mathcal{P},\ (i,j) \in \mathcal{K}_{od}^t,\ t \in \mathcal{T}.$$

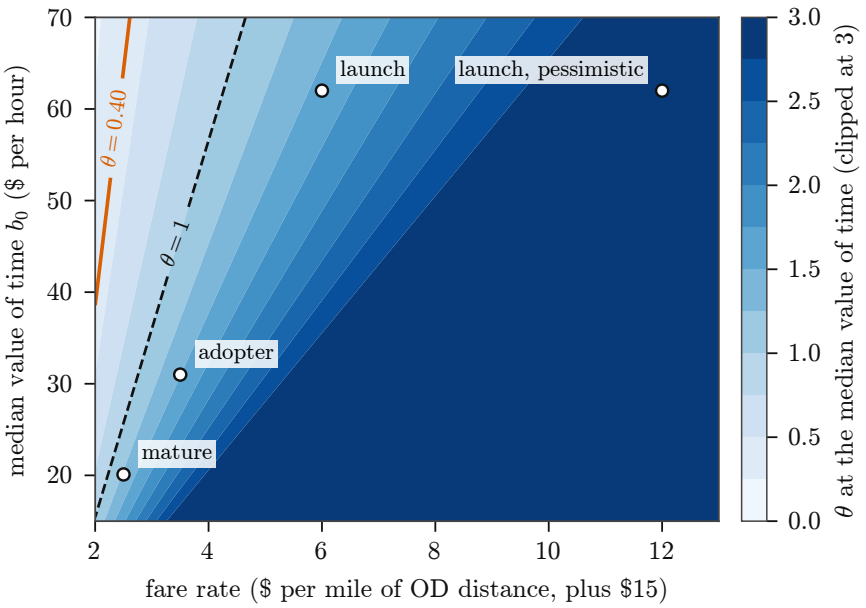


Figure 5: Required saving θ at the median value of time, over fare rates and values of time. The OD pair is illustrative: a 35-minute, $30 rideshare trip of 12 miles. Dots show the four scenarios of Table 4. θ is close to the paper's 0.40 only with low fares and high values of time. Most Chicago switchers therefore have a value of time above the median.

Option (o, i, j, d) then carries $D^t_{od}\, a^t_{oijd}\, x^t_{oijd}$ air-taxi passengers per day in period t. Of the D^t_{od} travelers, the share x is offered the option, and the share a of them accepts it.

7 Optimization models

7.1 Model P0: NYC validation (Table 4 of the paper)

The objective is passengers minus a cost γ per vertiport:

$$\max \quad \sum_{(o,d)\in\mathcal{P}} \sum_{(i,j)\in\mathcal{K}_{od}} D_{od}\, a_{oijd}\, x_{oijd} \;-\; \gamma \sum_{i\in\mathcal{V}} y_i \tag{21}$$

$$\text{s.t.} \quad \sum_{(i,j)\in\mathcal{K}_{od}} x_{oijd} \le 1 \qquad \forall (o,d)\in\mathcal{P}, \tag{22}$$

$$\sum_{j:(i,j)\in\mathcal{K}_{od}} x_{oijd} \le y_i \qquad \forall (o,d)\in\mathcal{P},\ i\in\mathcal{V}, \tag{23}$$

$$\sum_{i:(i,j)\in\mathcal{K}_{od}} x_{oijd} \le y_j \qquad \forall (o,d)\in\mathcal{P},\ j\in\mathcal{V}, \tag{24}$$

$$y_i \in \{0,1\}, \quad 0 \le x_{oijd} \le 1 \qquad \forall\, i,\ (o,i,j,d). \tag{25}$$

P0 has one period, so the index t is dropped. γ is measured in passengers. It is calibrated so that $\mu = 0.40$ gives the paper's three vertiports. μ is then reduced from 0.40 to 0.33 with γ fixed (Appendix A).

7.2 Model P1: one period (Chicago)

As in P0, the period index t is dropped. The objective is profit, or passengers with $\pi = 1$ and $g = 0$. Table 5 explains each constraint.

$$\max \quad \sum_{(o,d)\in\mathcal{P}} \sum_{(i,j)\in\mathcal{K}_{od}} \pi_{oijd}\, D_{od}\, a_{oijd}\, x_{oijd} \;-\; \sum_{i\in\mathcal{V}} g_i\, y_i \tag{26}$$

$$\text{s.t.} \quad \sum_{(i,j)\in\mathcal{K}_{od}} x_{oijd} \le 1 \qquad \forall (o,d)\in\mathcal{P}, \tag{27}$$

$$\sum_{j:(i,j)\in\mathcal{K}_{od}} x_{oijd} \le y_i \qquad \forall (o,d)\in\mathcal{P},\ i\in\mathcal{V}, \tag{28}$$

$$\sum_{i:(i,j)\in\mathcal{K}_{od}} x_{oijd} \le y_j \qquad \forall (o,d)\in\mathcal{P},\ j\in\mathcal{V}, \tag{29}$$

$$\sum_{i\in\mathcal{V}} y_i \le p, \tag{30}$$

$$\frac{1}{H} \sum_{(o,d)\in\mathcal{P}} \sum_{(i',j')\in\mathcal{K}_{od}:\ i\in\{i',j'\}} D_{od}\, a_{oi'j'd}\, x_{oi'j'd} \;\le\; Q\, y_i \qquad \forall i\in\mathcal{V}, \tag{31}$$

$$y_i \in \{0,1\}, \quad 0 \le x_{oijd} \le 1 \qquad \forall\, i,\ (o,i,j,d). \tag{32}$$

Table 5: Meaning of the constraints of Model P1.

Constraint	Meaning
(27) Assignment	The shares of an OD pair add up to at most 1. The rest stays on the ground
(28) Departure link	Travelers can only board at a vertiport that is built
(29) Arrival link	Travelers can only land at a vertiport that is built
(30) Budget	At most p vertiports are built. All reported runs use it, with $p = 1, \ldots, 10$
(31) Capacity	At site i, departing plus arriving passengers per hour are at most Q. H is the length of the period in hours. Optional (Section 7.4)

7.3 Model P2: one network for all periods (RQ2)

P2 chooses one network y for all periods. The routing x^t can change from one period to the next:

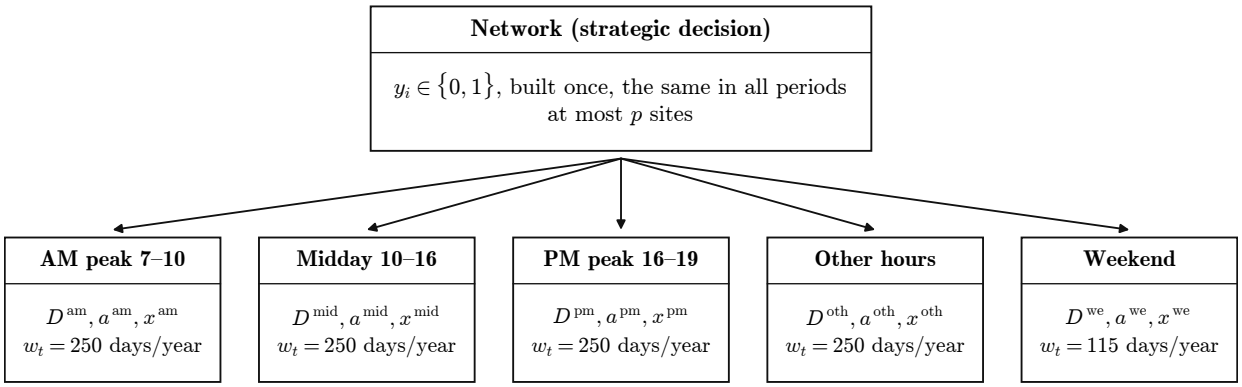
$$\max \quad \sum_{t\in\mathcal{T}} w_t \sum_{(o,d)\in\mathcal{P}} \sum_{(i,j)\in\mathcal{K}^t_{od}} \pi^t_{oijd}\, D^t_{od}\, a^t_{oijd}\, x^t_{oijd} \;-\; G \sum_{i\in\mathcal{V}} g_i\, y_i \tag{33}$$

$$\text{s.t.} \quad \sum_{(i,j)\in\mathcal{K}^t_{od}} x^t_{oijd} \le 1 \qquad \forall (o,d)\in\mathcal{P},\ t\in\mathcal{T}, \tag{34}$$

$$\sum_{j} x^t_{oijd} \le y_i, \qquad \sum_{i} x^t_{oijd} \le y_j \qquad \forall (o,d)\in\mathcal{P},\ i,j\in\mathcal{V},\ t\in\mathcal{T}, \tag{35}$$

$$y_i \in \{0,1\}, \quad 0 \le x^t_{oijd} \le 1 \qquad \forall\, i,\ (o,i,j,d),\ t\in\mathcal{T}. \tag{36}$$

w_t is the number of days per year of period type t. $G = 365$ is the number of days per year. Both terms are therefore yearly. In the reported runs, $\pi = 1$ and $g = 0$, and the budget (30) is added. The objective is then yearly passengers. Figure 6 shows the structure. The capacity constraint (31) can be added for each period (Section 7.4).



Robust plan: $y^R = \arg\max_y \sum_t w_t \, \Pi_t(y)$. Plan for one period t': $y^*_{t'} = \arg\max_y \Pi_{t'}(y)$.

Regret of planning for t': $\text{Reg}(t') = 1 - \sum_t w_t \, \Pi_t(y^*_{t'}) \,/\, \sum_t w_t \, \Pi_t(y^R)$.

Figure 6: Structure of Model P2 and comparison with single-period plans (passenger objective).

7.4 Vertiport capacity

In P2, constraint (31) is applied in every period. The period lengths are $H_t = 3$, 6, 3, 12 and 24 hours (AM, midday, PM, other weekday hours, weekend). Demand is spread evenly within each period. Capacity takes the values $Q \in \{\infty, 200, 100, 50\}$ passengers per hour (departing plus arriving). These values of Q are illustrative.

8 Model properties

Proposition 1 (The routing is integral for a fixed network)**.** *For any fixed $y \in \{0,1\}^{\mathcal{V}}$, Model P1 without capacity has an optimal solution in which each OD pair uses at most one vertiport pair, with $x_{oijd} \in \{0,1\}$. The optimal routing sends OD pair (o,d) through*

$$(i^*, j^*) \in \arg\max\big\{\pi_{oijd}\, a_{oijd} : (i,j) \in \mathcal{K}_{od},\ y_i = y_j = 1\big\}$$

if that maximum is positive, and keeps it on the ground otherwise.

Sketch. With y fixed, the problem separates by OD pair. If $y_i = 0$ or $y_j = 0$, the linking constraints force $x_{oijd} = 0$. Otherwise, the assignment constraint already implies them. The remaining set $\{x \geq 0 : \sum x \leq 1\}$ is a simplex. A linear objective reaches its maximum at a vertex of this set: all zeros or one option at 1. □

Remark. Proposition 1 gives the value of a network in closed form:

$$\Pi(y) = \sum_{(o,d)} D_{od} \, \max\Big\{0, \max_{(i,j)\in\mathcal{K}_{od}:\, y_i = y_j = 1} \pi_{oijd}\, a_{oijd}\Big\} - \sum_i g_i y_i.$$

In words, each OD pair uses its best open option, if that option is better than staying on the ground. With this formula, any network can be evaluated quickly under any demand model. The metrics of Section 9 need this.

Proposition 2 (Strength of the linking constraints)**.** *The aggregated linking constraint $\sum_j x_{oijd} \leq y_i$ implies $x_{oijd} \leq y_i$ for every j. The reverse is not true. The formulation is therefore at least as tight as the classical "strong" facility-location formulation.*

Structure and size. In P1, each customer (OD pair) needs *two* open facilities. This makes P1 closer to hub location (Campbell 1994; Alumur and Kara 2008) than to uncapacitated facility location (Cornuéjols, Nemhauser and Wolsey 1990). The main Chicago instance has 16 binaries and 2,289 options over all periods ($s \geq 0$). Of these, 617 weekday options have $s \geq 0.15$. The largest robustness instance has 77 binaries and about 36,000 options. Gurobi solves each instance in seconds.

9 Evaluation metrics

Let $\Pi^{\text{het}}(y)$ be the value of network y when each OD pair has its own expectations. Let $\Pi^{\text{unif}}(y)$ be its value under the uniform benchmark of Section 5. In both cases the routing is optimal (Proposition 1).

Definition 1 (Value of heterogeneity, RQ1)**.** Let y^{het} be optimal with OD-specific shares. Let y^{unif} be optimal with the pooled shares of Section 5. We take the OD-specific model as the truth:

$$\text{VOH} = \Pi^{\text{het}}(y^{\text{het}}) - \Pi^{\text{het}}(y^{\text{unif}}) \ \geq 0, \qquad \text{VOH\%} = \frac{\text{VOH}}{\Pi^{\text{het}}(y^{\text{het}})}.$$

VOH is the loss from planning with one average expectation.

Definition 2 (Forecast error of the average-customer model, RQ1)**.** Take the network y^{unif}. The forecast error FE is the relative gap between what the pooled model predicts and what the OD-specific model gives: $\text{FE} = \big(\Pi^{\text{unif}}(y^{\text{unif}}) - \Pi^{\text{het}}(y^{\text{unif}})\big)/\Pi^{\text{het}}(y^{\text{unif}})$. A negative value means the pooled model predicts too few passengers.

Definition 3 (Regret of a single-period plan, RQ2)**.** Let y^R solve P2. Let $y^*_{t'}$ solve P1 with the demand of period t' only. Both use the same budget p. The regret of planning for t' is the share of yearly passengers lost:

$$\text{Reg}(t') = 1 - \frac{\sum_t w_t \, \Pi_t(y^*_{t'})}{\sum_t w_t \, \Pi_t(y^R)} \ \geq 0.$$

$\Pi_t(y)$ is the value of network y in period t. With capacity, it is computed by re-solving the routing with y fixed.

Network similarity. We compare two networks with site sets A and A' in two ways. The first is the Jaccard index $|A \cap A'|/|A \cup A'|$: the share of sites they have in common. The second is, for each site in A', the distance to the nearest site in A. We use both to compare with the paper and with Volakakis and Mahmassani (2024).

Benchmark: p-median (not run yet). This benchmark ignores the air-taxi choice. It chooses p sites that minimize the demand-weighted distance from each origin to its nearest site. With $D_o = \sum_d D_{od}$ the number of trips that start in o,

$$\min \sum_o \sum_i D_o \, \ell_{oi} \, z_{oi} \quad \text{s.t.} \ \sum_i z_{oi} = 1, \ z_{oi} \leq y_i, \ \sum_i y_i = p, \ y \in \{0,1\}, \ z \geq 0.$$

Its network is then evaluated with $\Pi(y)$. The paper reports 22–53% lower profit for this benchmark.

10 Robustness checks

10.1 Exact evaluation of all networks

Without capacity, Proposition 1 gives the value of every network in closed form. With 16 candidates and $p \leq 8$, there are 39,186 networks. All of them are evaluated directly. This reproduces the Gurobi solutions of P1 and P2 exactly. It also makes the checks below fast, because no solver is needed.

10.2 Day-level bootstrap

The bootstrap measures how much the results depend on which days were observed (Efron and Tibshirani 1993). The 22 weekdays and 9 weekend days of March 2026 are resampled with replacement 200 times. Weekdays and weekend days are resampled separately. Days are resampled, not trips, because trips on the same day share traffic conditions. For each resample, demand, ground times, options and shares are recomputed. Every network is then evaluated again. Reported intervals are the 5th and 95th percentiles. They measure sampling variation in March-like conditions. They do not measure uncertainty in the assumed parameters.

10.3 Seasonal check

The same pipeline is applied to July 2025 and January 2026. For each p, the network planned for one month is evaluated on the other two. An all-season network maximizes the sum of the three monthly values. Each monthly value is divided by its own optimum, so all months have equal weight.

10.4 Transfer penalty

The penalty h of Section 4.3 takes the values 0, 5, 10 and 15 minutes, each with $s_{\min} = 0.15$ and $s_{\min} = 0$. Flight overhead and pooling wait also add fixed minutes to τ. Cruise speed has a similar effect. These tests therefore cover them too.

11 Relation to the original model

Table 6 compares this model with the paper. The two extensions of this project are in bold.

Table 6: Comparison with Wang, Jacquillat and Vaze (2022).

Component	**Wang, Jacquillat and Vaze (2022)**	**This project**
Vertiport decisions	location and capacity level z_{ih}, fleet Γ	location y_i (capacity optional)
Matching	y_{oijd} binary, π_{oijd} fractions	x^t_{oijd} (integral at optimum)
Pooling wait	endogenous, $\delta_{ij} \geq (n-1)/\chi_{ij}$ (second-order cone)	fixed $\bar{\delta}$
Detours, reliability	endogenous (queueing network)	not modeled
Fleet, repositioning, charging	modeled	not modeled
Demand function	DRO or multinomial logit S-curve, inside the model	exact share (main) and DRO S-curve (comparison), both precomputed; optional transfer penalty h
Customer expectations	uniform μ (0.40), σ	**uniform and OD-specific, from fares (RQ1)**
Time of day	one demand snapshot (morning commute)	**five periods, robust network (RQ2)**
Solution method	adaptive discretization	MILP, Gurobi

The simplified model is close to the paper's "exogenous operations" benchmark. A natural extension is to make the pooling wait endogenous again. Wang et al. (2022, Lemma 1) show how to write it as a second-order cone constraint.

12 Runs reported

Table 7 lists the runs behind the reported results.

Table 7: Runs reported. "Enumeration" means the exact evaluation of Section 10.1.

Run	Content	Model
V1	NYC validation: μ reduced from 0.40 to 0.33 with a calibrated cost γ per vertiport (Appendix A)	P0
C1	RQ1: OD-specific vs pooled shares, four fare scenarios, five periods and weekday, $p = 1, \ldots, 10$; 16 sites and all 77 areas ($k = 3$ and $k = 8$)	P1
C2	RQ2: robust network vs single-period networks, three scenarios, $p = 2, \ldots, 8$; 16 sites and 77 areas	P2, P1
C3	Capacity $Q \in \{\infty, 200, 100, 50\}$, adopter scenario	P2
C4	Day-level bootstrap, 200 resamples, three scenarios	enumeration
C5	Transfer penalty $h \in \{0, 5, 10, 15\}$, $s_{\min} \in \{0.15, 0\}$	enumeration
C6	Seasonal check: July 2025, January 2026, March 2026	enumeration

Not yet run: p-median benchmark; spread of the value of time (CV_b from 0.5 to 1.2); March 2019 comparison; endogenous pooling.

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A NYC validation results (Model P0)

The data are NYC yellow taxi trips of March 2019, weekdays 7–10 am. Demand is multiplied by 2, as in the paper. There are 26 candidate zones. Other values: $v = 240$ km/h, $T_{\mathrm{oh}} = 10$, $\bar{\delta} = 5$,

$\sigma = 0.20$, and $\gamma = 36.75$ passengers. Any γ between 32 and 41.5 gives 3 vertiports at $\mu = 0.40$. Table 8 gives the results.

Table 8: NYC validation: number of sites as μ decreases, compared with Table 4 of the paper.

μ	**Sites (ours)**	**Sites (paper)**	**Passengers**	**Change in our network**
0.40	3	3	125.6	JFK, Times Sq/Theatre District, Gramercy
0.39	3	3	136.6	–
0.38	4	3	187.0	+ Upper East Side North
0.37	4	4	201.9	–
0.36	6	4	290.3	+ Kips Bay, LaGuardia, TriBeCa; − Gramercy
0.35	6	7	316.9	–
0.34	7	9	382.8	+ Gramercy, Midtown Center; − Kips Bay
0.33	7	10	416.7	–

The number of vertiports follows the paper closely. Most site differences are adjacent zones, for example Times Sq and Clinton East. The model puts the LaGuardia-area vertiport at the airport. The paper puts it in Astoria for operational reasons: shorter flights, higher reliability and shorter detours (Section 7.1.1 of the paper). This difference shows the effect of the operations layer, which this model does not include.